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Assessing the Supply–Demand Matching and Spatial Flow of Urban Cultural Ecosystem Services: Based on Geospatial Data and User Interaction Data

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Abstract: Cultural ecosystem services (CESs) reflect the interaction between ecosystems and human well-being. Owing to constraints in data availability and existing methodological limitations, deriving information from non-material ecosystem attributes was inadequate. We took Yulin City, located in the northern Shaanxi Loess Plateau, as a case study. Based on open-source geospatial data and user interaction data from social media, a coupled multi-source model was applied to elucidate the spatial distribution of CESs' supply–demand flow. The Maxent and LDA model were utilized to quantify CES supply–demand, whereas the breakpoint and gravity model were applied to explain the direction and intensity of CES flow. The results indicated the following: (1) aesthetic was the most perceivable CES in Yulin, with 27% high supply areas and four demand topics. And the perception of the educational CES was the least pronounced, with only 2% of high supply areas and two demand topics. (2) Yulin exhibited a notable mismatching in CES supply–demand, with the supply–demand matching area constituting only approximately 10%. In the center of the city, CESs displayed a spatial pattern of a supply–demand deficit, while areas farther from the city center presented a spatial pattern of a supply–demand surplus. (3) The flow of CESs followed a pattern of movement from peripheral counties to central counties and from less developed counties to more developed counties. We proposed the following targeted recommendations: introducing low-perception CESs to promote the enhancement of ecosystem services (ESs); and alleviating CES supply–demand mismatches by enhancing transportation accessibility and protecting the ecological environment. Simultaneously, attention should be directed towards the developmental disparities between counties, providing differentiated guidance for CES spatial flow. Our study provided a theoretical foundation for understanding CES supply–demand flow and offered scientific insights for the spatial development of urban CES.

Keywords: cultural ecosystem services; user interaction data; supply–demand matching; spatial flow; the LDA model



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1. Introduction

Cultural ecosystem services (CESs) refer to the non-material benefits humans derive from ecosystems through spiritual enrichment, cognitive development, and aesthetic experience [1]. Human subjective emotions and preferences are integrated into the assessment framework, reflecting the socio-cultural attributes of ecosystems [2]. However, due to their intangible and subjective nature, which reflects human spiritual values, they have been inherently difficult to quantify [3]. In discussions on ESs, the quantitative assessment of

CESs has often been overlooked, as scholars primarily focused on material services such as water production, carbon storage, and soil conservation. This obstructed the incorporation of CESs into the decision-making processes [4]. But in socio-ecological systems, CESs make multiple contributions to enhancing human well-being, promoting sustainable economic development and maintaining ecosystem integrity [5,6]. Urban cultural ecosystem services link complex and dynamic urban areas with the intangible cultural attributes of ecosystems; there is a need for adequate integration of CESs within urban planning.

With rapid socio-economic development, the connection between ecosystems and diverse cultures became increasingly weakened. CES supply–demand exhibited significant spatial disparities and imbalances, resulting in varying intensities of CES spatial flow [7,8]. CES supply refers to the services and actual products that ecosystems can provide [9], while demand refers to the CESs that regional beneficiaries require or anticipate [10]. The spatial movement of elements between surplus and deficit areas forms the CES flow [11]. The realization of CES value can be examined from the supply–demand–flow perspective in the context of sustainable development goals. This provides a more comprehensive framework for urban landscape management decisions and policy formulation [12,13]. Traditional CES assessments primarily used indicator-based value estimation and questionnaire or interview methods based on preferences. Indicator-based value estimation utilized existing data to assess supply and demand levels of CESs [14]. Previous studies revealed CES supply–demand matching patterns by comparing green space area with population density [15], while CES utilization costs were quantified by counting the ticket or travel expenses [16]. However, these previous studies focused on specific types of CESs, making it difficult to capture intangible and non-material CESs. Questionnaires and interviews were the primary methods for evaluating CESs [17,18]. They involved collecting preferences and opinions from residents, tourists, and experts to assign scores or frequencies; the social values for ecosystem services (SolVES) model provided technical support for such analyses [19]. This approach addressed non-material aspects of CESs but required substantial time and effort [20]. Although the soil and water assessment tool (SWAT) [21], the multi-region input–output (MRIO) model [22], the service path attribution networks (SPANs) [23], and Bayesian network models (BBN) [24] developed various distributed ecological models for ecosystem service flow (ESF) research, these methods often relied on large datasets and were primarily applied to specific types of ESF simulations. The evaluation of CES flow still faced challenges such as limited data accessibility, unclear methodological tools, and restricted research applications. A comprehensive approach is needed to conduct a detailed assessment of CES supply–demand–flow characteristics.

As a form of social sensing data, real-time, geotagged social media data provide an opportunity to explore human perceptions and experiences of CESs [25,26]. For example, Wang et al. quantified the practical utilization of CESs by measuring the number of tourists through social media [27]. Yoshimura et al. and Sinclair et al. used photo density from open websites like Flickr and Panoramio to quantify the perceptual experience and supply–demand indicators of CESs [28,29]. Inácio et al. applied online reviews to map the aesthetic and recreational value [30]. Compared to surveys and interviews, user interaction data from social media are both faster and more cost-effective, significantly reducing the cost of CES assessment, while more accurately and effectively capturing the spatial flow of CESs [31]. However, previous studies have focused on statistical analysis of the number of photos or documents, failing to establish the topic connections behind the semantics of the texts [10,32]. It is important to explore the connections within comments' text to study the spatial interactions among various CESs. Our study, grounded in open-source geospatial data and large-scale user interaction data, quantified the spatial distribution of CES supply–demand flow by integrating multiple models. The machine learning-based maximum

entropy (Maxent) model was initially applied to species distribution prediction [33]. Under CES geolocation conditions, it not only enables the spatial representation of multiple types of CES supply but also captures the response relationship between environmental factors and CES supply [34,35]. As natural language processing evolves, the topic model for text classification has been widely applied to analyze beneficiaries' perceived demand [36]. It was commonly employed for text information mining and semantic topic analysis [37], enabling the quantification of CES demand. Moreover, by leveraging the spatial benefit flow index, the breakpoint model, and gravity model, the interactions and transfer intensities of CESs between different regions can be quantified, providing important references for spatial simulation of CES flow [38,39].

Yulin City is in the central region of the Loess Plateau, characterized by severe soil erosion and ecological fragility. Bordering the Inner Mongolia Plateau, Yulin exhibits a unique blend of agricultural and nomadic cultures, reflecting the folk traditions of northern Shaanxi. As a result of the interplay of various intersecting factors, Yulin has been undergoing rapid shifts in land use, accompanied by an increasingly sensitive ecological environment and the decline of its cultural and tourism resources [40]. Identifying the characteristics of CES supply–demand flow is imperative to reconcile the tension between ecological protection and social development. This study used Yulin City as a case study and developed an assessment framework for urban CES supply–demand flow. The specific objectives included the following: (1) Establish a theoretical framework for CESs, applying the Maxent and Latent Dirichlet Allocation (LDA) model to evaluate the CES supply–demand levels, and identify the spatial patterns for matching CES supply–demand. (2) Incorporate the breakpoint model and gravity model to explain the direction and intensity of CES flows, illustrating the spatial transfer mechanisms of CESs. (3) Provide recommendations for the spatial optimization of CESs from the perspectives of perceptual experience, supply–demand matching, and spatial flow. Our study integrated social media data throughout the entire CES research process. By leveraging user interaction data to interpret perceptual differences in CESs, this approach overcame the data limitations inherent in traditional survey methods. Coupled multi-source models to achieve a comprehensive evaluation of CES supply–demand flow, elucidating the flow mechanisms of CESs between deficit and surplus areas. The results established a scientific foundation for landscape planning and resource allocation in cities and larger regions.

2. Materials and Methods

2.1. Study Area

Yulin City (36°57' N–39°35' N, 107°28' E–111°15' E) is located in northern Shaanxi Province, China. It covers a total area of 42,920.2 km² and consists of 12 administrative districts, including 2 districts, 1 county-level city, and 9 counties (Figure 1). The permanent resident population in 2023 was 3.61 million. The city experiences an annual average temperature of 8 °C, a mean annual precipitation of approximately 400 mm, and an average altitude of 1220 m. The terrain is elevated in the northwest and diminishes in the northeast, featuring diverse and complex landforms.

It is a critical region in China for ecological environmental construction and the “Grain for Green” program. Following decades of restoration effort, the forest coverage rate in Yulin City has risen from an initial 0.9% to 36% [41,42]. As a national historical and cultural city, Yulin is not only the birthplace of Loess culture but also holds the heritage and development of Yellow River civilization [43]. In the context of ongoing urban and rural expansion and the increasing demand for cultural and recreational activities, examining the conflicting relationship between the ecological environment and human society, as

well as evaluating the supply–demand matching and spatial flow of CESs, is essential for ecological conservation and sustainable development in Yulin City.

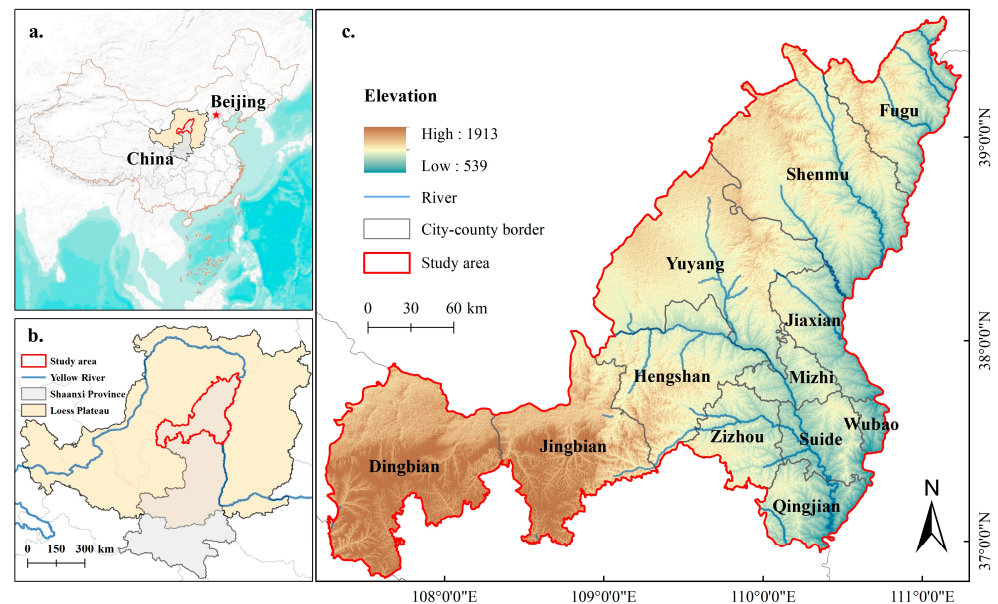


Figure 1. Location of Yulin City. (a) Location of Yulin City in China, (b) location of Yulin City in the Loess Plateau and Shaanxi Province, and (c) boundaries of Yulin City and elevation map.

2.2. Data Collection

This study primarily utilized geospatial data and user interaction data. We collected four natural factors and three spatial distance factors as environmental factors for the Maxent model (Figure A1). All environmental factors were tested for correlation (Figure A2). User interaction data included CES supply points and comments' text. Using the Gaode Map API services, we filtered Points of Interest (POIs) with CES functions, such as parks, scenic areas, and squares [44]. Meanwhile, we collected comments' text from the "Nearby Travel" section of the site. Additionally, we collected per capita GDP data for each county to compare the relationship between CES spatial flow and economic levels. Table 1 shown details of data description and sources.

Table 1. Data description and sources.

Data	Description	Units	Year	Source
Geospatial data	LU: land use	/	2022	https://zenodo.org/records/12779975 (accessed on 10 December 2024)
	Slope: terrain slope	degree	2022	https://www.gscloud.cn/ (accessed on 10 December 2024)
	DEM: digital elevation model	m	2022	https://www.gscloud.cn/ (accessed on 10 December 2024)
	NDVI: normalized difference vegetation index	/	2022	http://www.nesdc.org.cn/ (accessed on 1 August 2024)
	DicW: distance from water	m	2022	https://www.openstreetmap.org (accessed on 20 October 2024)
	DisR: distance from road	m	2022	https://www.openstreetmap.org (accessed on 20 October 2024)
	DisT: distance from town	m	2022	https://lbs.amap.com/ (accessed on 1 December 2023)

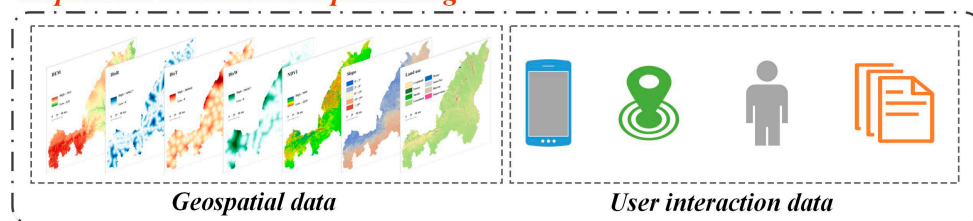
Table 1. Cont.

Data	Description	Units	Year	Source
User interaction data	POI with CES functions, including ID, latitude, longitude, and tags	/	2023	https://lbs.amap.com/ (accessed on 1 December 2023)
	Website consumer comments text, including ID, latitude, longitude, time, ratings, and reviews	/	2023	https://www.dianping.com/ (accessed on 1 December 2023)
Per capita GDP data	Gross Domestic Product Per Capita	/	2022	https://tjj.yl.gov.cn/tjsj/tjgb/ (accessed on 25 March 2025)

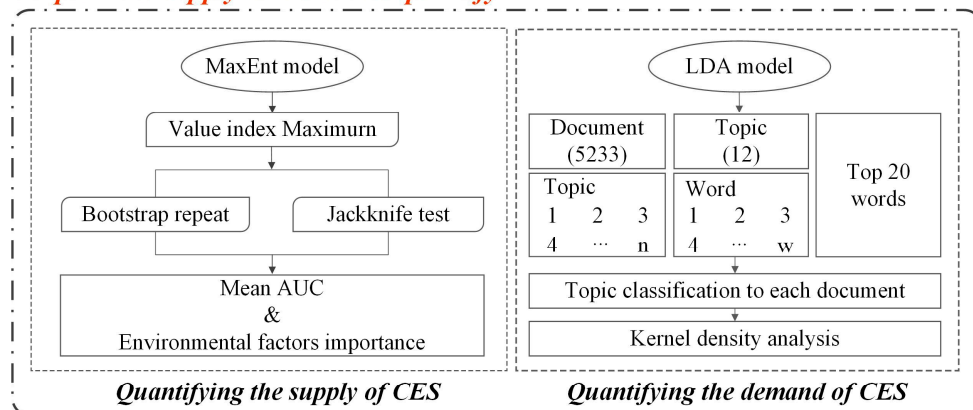
2.3. Methods of CES Quantification

Our research framework is shown in Figure 2. First, we used geospatial data and user interaction data to establish a database for CES assessment. Secondly, the Maxent model and LDA model were applied to quantify CES supply and demand, identifying the perceptual differences across various categories of CES. Thirdly, we evaluated the spatial distribution of CES supply–demand matching, quantifying areas of concentrated supply–demand and determining the surplus and deficits of CESs. Finally, the breakpoint model and gravity model were used to assess CES flow direction and intensity, simulating the spatial transfer of CESs.

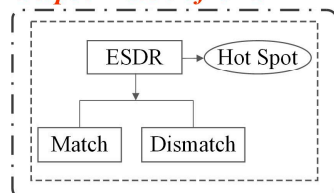
Step1. Data collection and processing



Step2. CES supply and demand quantify



Step3. Match of CES



Step4. CES supply-demand-flow characteristics

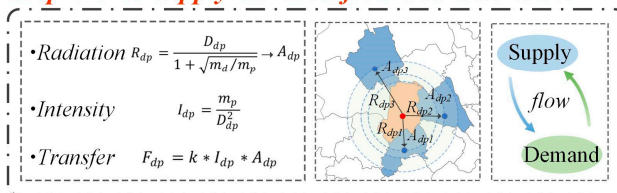


Figure 2. Framework for CES supply–demand–flow assessment.

In the past, research on CESs primarily concentrated on aesthetic or recreational services [45–47], with less emphasis placed on spiritual and educational services. Our study

aimed to address this gap by employing CES definitions from the MEA and TEEB reports, while incorporating regional cultural characteristics [1,48]. We quantified and mapped CESs based on four aspects: aesthetic, recreational, spiritual, and educational (Table 2).

Table 2. CES types and attributes.

Types	Meaning	Tags	Keywords
Aesthetic	Natural landscapes with ornamental value and aesthetic enjoyment exemplify the ecological service value derived from the harmonious coexistence between humans and nature.	Tourist attractions; Natural place name	Scenic spot, pavilion, forest, pine, bamboo, apricot, plum, cliff, peak, valley, attraction; mountain, water, river, stream, lake, sea, ocean, pond, pool, waterfalls, shoal beach, wetland
Recreational	The capacity for recreational activities within the natural ecosystem reflects the values and needs tied to folklore and cultural traditions.	Recreation place; Holiday and Nursing resort	Leisure, square, camp, base, garden, village, slide, street; vacations, estates, hot springs
Spiritual	A place with spiritual or religious significance reflects the profound cultural and historical importance to humanity.	Buddhist and Taoist temple; Church; Mosque	Confucian, Buddhist, and Taoist temples, filial piety, virtue, ancient, history,
Educational	Sites designated for natural science research and educational activities.	Memorial hall; Science/Culture and Education service	Memorials, culture, humanities, auditoriums, pavilions, palaces, gardens, science, education

2.3.1. Quantifying the Supply and Demand of CESs

(1) Quantifying CES supply

Based on Gaode API's "POI Classification Code Table" and previous studies, we established the tags and keywords for CESs [49,50]. After filtering and removing duplicate entries, we identified 1418 CES supply points, including 836 for aesthetic, 136 for recreation, 196 for spiritual, and 250 for educational services. The Maxent model, as a machine learning algorithm, can predict the distribution characteristics of features and establish intricate nonlinear relationships between environmental factors and changes in the target variable [51]. The model was initially applied to assess the potential for species distribution [52]. With the support of machine learning and other algorithms, its applications have broadened to fields such as urban expansion [53], settlement pattern optimization [54], and land use change [55]. We applied it to the assessment of CES supply capacity. The formula is as follows:

$$H(X/Y) = -\sum_i^n p(X,Y) \log_p(X,Y) \quad (1)$$

$$X^* = \operatorname{argmax} H(X/Y) \quad (2)$$

where X is the distribution of CES supply points, Y is the environmental factors, n is the number of training samples, $p(X,Y)$ is the probability function, $H(X/Y)$ is the information entropy of the training samples, and X^* is the value of X at which the entropy $H(X/Y)$ reaches its maximum.

We utilized the "Maxent 3.4.4" Java environment developed and updated by Phillips et al. [56]. Environmental factors and CES supply points were imported into the software. The 75% dataset was selected as training data to ensure sufficient samples for evaluating the nonlinear relationships between environmental variables and CES supply, while the remaining 25% served as the testing set to maintain sufficient data volume for rigorous validation [56,57]. The supply result was established by the average potential value after

ten repetitions to ensure the model's stability [58]. The model's feasibility was validated by averaging the Area Under the Curve (AUC) values. The AUC assesses the model's performance by comparing the scores of different samples, regardless of specific classification thresholds [59,60]. The mean AUC for the aesthetic, recreational, spiritual, and educational services was 0.649, 0.913, 0.853, and 0.940. All these values exceeded the random distribution simulation value of 0.5, indicating that the model performed well in Yulin City, and its evaluation results were reliable and robust. Therefore, it could be used to explain the spatial distributions of CES supply and its driving factors [28].

(2) Quantifying CES demand

In practice, interactions exist between different CES types, and the supply and demand for these types are not single. For example, a park can provide both aesthetic and recreational services. The LDA model is one of the most used topic models [61]. It identifies relevant topics from documents by establishing a three-tier hypothesis involving documents, topics, and words, positing that each document contains multiple topics, with each topic consisting of several words [36,61]. The LDA model determines the probability distribution of different topics within documents. By categorizing topics into distinct CES types, the probability of different CESs can be identified (Figure 3).

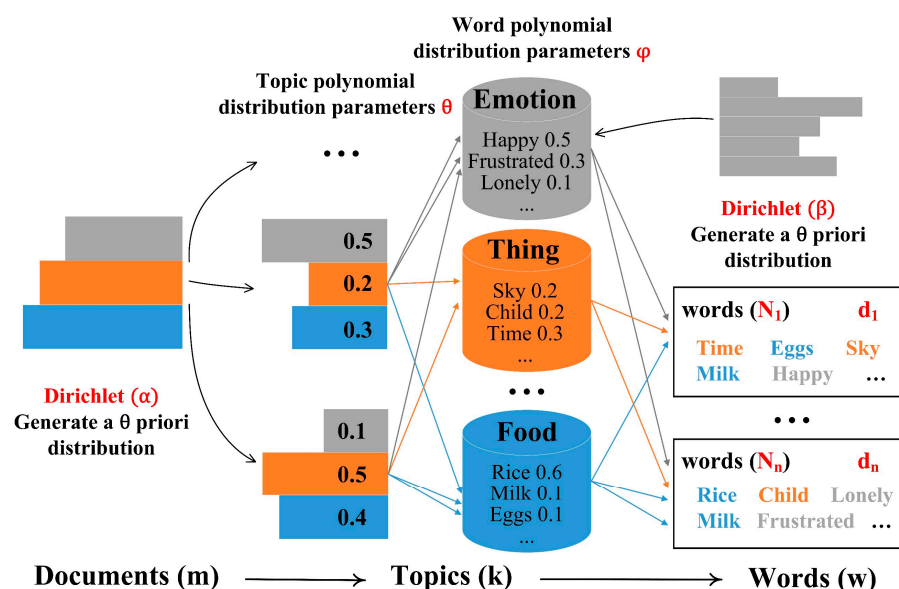


Figure 3. The Latent Dirichlet Allocation model.

It uses the Gibbs sampling method to iteratively infer parameters, with the joint probability distribution expressed as follows:

$$P(w, z, \theta, \varphi | \alpha, \beta) = P(\theta | \alpha) \left(\prod_{k=1}^K P(\varphi_k | \beta) \right) \prod_{d=1}^M \prod_{n=1}^{N_d} P(z_{d,n} | \theta_d) P(w_{d,n} | \varphi_{z_{d,n}}) \quad (3)$$

where α and β are Dirichlet prior parameters—we set these parameters to 0.05 and $0.01 \cdot \theta$ is the document–topic probability distribution, φ is the topic–word probability distribution, z is the topic label to words, w is the word label, K is the number of topics, M is the number of documents, d is the document label, N_d is the number of words in document d , θ_d is the topic distribution of document d , φ_k is the word distribution of topic k , $z_{d,n}$ is the topic label for the n -th word in document d , and $w_{d,n}$ is the n -th word in document d . The number of topics, where the perplexity reduction rate plateaued, was used as the value for parameter K . After 2000 iterations, the optimal number of topics was 12 (Figure A3) [62].

As of 1 December 2023, we had collected 5233 comment texts (i.e., documents) from Dianping, selecting sites with more than three reviews. With python 3.12 support, we used the jieba package for word segmentation and filtering. Nouns, verbs, and adjectives with actual meaning were retained, and prepositions, pronouns, and conjunctions without specific meaning were excluded. The LAD model was used to identify topics and feature words. Latent topics express specific attributes of a particular category of CESs. Considering the meanings of high-frequency words in the topics and classifying them with the definitions of different CESs in Table 2, a matching system was established between CESs and topics. Finally, we quantified the CES demand value using the following formula and generated a demand map through kernel density analysis:

$$S_{nc} = \sum \theta_{nc} \quad (4)$$

θ represents the topic–document probability distribution, n denotes the various sites, and c refers to the same CES category.

2.3.2. Identifying the Supply–Demand Matching of CES

The ecosystem supply–demand ratio (*ESDR*) links the supply and demand of CESs [63], showing whether there is a surplus, balance, or deficit [64,65]. By standardizing the supply–demand values, we utilized the *ESDR* to assess how well they match or mismatch. The formula is as follows:

$$ESDR = 2 * (S - D) / (S_{max} + D_{max}) \quad (5)$$

where S and D are the normalized values of CES supply and demand, and S_{max} and D_{max} are the maximum normalized values. The *ESDR* ranges from -1 to 1 . A positive *ESDR* value indicates that the CES supply exceeds the demand, while a negative *ESDR* value signifies that the CES supply falls short of the demand. When the *ESDR* is approximately 0 , it reflects an equilibrium between the CES supply and demand.

Ecosystem services tend to spill over into surrounding areas after fulfilling local demand, prioritizing compensation for nearby regions and forming geographic clusters [66]. Therefore, not all surplus and deficit areas can be considered as supply sources and demand sinks. We used the Getis-Ord G_i^* to exclude spatially dispersed units and determine the spatial aggregation of CES supply–demand:

$$G_i^* = \frac{\sum_{j=1}^n w_{ij}(d)x_j - \bar{X}\sum_{j=1}^n w_{ij}(d)}{S\sqrt{\frac{n\sum_{j=1}^n w_{ij}^2(d) - \left(\sum_{j=1}^n w_{ij}(d)\right)^2}{n-1}}} \quad (6)$$

where n is the number of spatial features; w_{ij} is the distance between features i and j ; x_j is the value of the *ESDR*; $\bar{X}$ is the mean value of the *ESDR*. The hotspot areas were recognized as CES supply surplus areas, while the coldspot areas were designated as CES demand compensation areas [66,67]. Due to the limited number of county-level units, conducting a spatial analysis of CESs proved challenging. To express the spatial transfer of CESs more accurately, we conducted a visualization analysis of CES flow at the township scale [68].

2.3.3. Evaluating the Spatial Flow of CESs

Owing to the influence of human willingness to pay, distance costs emerged as the primary constraint on CES utilization [69]. The matching of CES flow between different areas depended not only on supply–demand differences but also on spatial distance [70]. The breakpoint model and gravity model were commonly used to analyze urban radiation effects, reflecting a city’s influence on surrounding areas based on development intensity and distance [71]. Similar to the theory of ESF, these models had been successfully applied

in studies of ESF across various regions [38,72]. We used the breakpoint model to measure the geographical transference scope of CES flow and applied the gravity model to assess the radiation capacity and transfer intensity from the surplus areas to the deficit area [73,74]. The calculation process was as follows:

Firstly, the radial distance from the surplus area to the deficit area was calculated:

$$R_{dp} = \frac{D_{dp}}{1 + \sqrt{m_d/m_p}} \quad (7)$$

where R_{dp} represents the CES flow radius, defined as the distance from the center of the surplus area to the breakpoint. D_{dp} is the center distance from the surplus area p to the deficit area d . m_p and m_d represent the CES supply value of the area p and d . A buffer analysis with R_{dp} as the radius determines the radiation range A_{dp} of the surplus area.

Secondly, the radiation intensity of the surplus area was determined by employing the gravity model:

$$I_{dp} = \frac{m_p}{D_{dp}^2} \quad (8)$$

where I_{dp} represented the field intensity from the surplus area p to the deficit area d , indicating the flow intensity. In this study, the field intensity values were magnified by a factor of 10,000 for enhanced clarity, without affecting the comparative results.

Finally, based on the impact area and radiation range, we determined the transfer intensity of CES between each surplus and deficit area.

$$F_{dp} = k * I_{dp} * A_{dp} \quad (9)$$

where F_{dp} represents the CES transfer volume from the surplus area p to the deficit area d . The coefficient k is a spatial conversion factor influenced by terrain and the movement of natural resources. Considering the socioeconomic attributes of CES and referring to existing studies, k was set to 0.4 [75].

3. Results

3.1. Supply and Demand Characteristics of CESs

3.1.1. Characteristics of CES Supply

From a holistic spatial perspective, the city center served as a pivotal hub for urban activities and information exchange, exhibiting a concentration of diverse CESs and high (high and very high) supply levels. There was also a trend of high CES supply along urban arterial roads, forming a linear distribution pattern (Figure 4). Aesthetic services had the highest supply capacity, with 27% classified as high supply and 38% as medium supply. Forest and ecological resources exerted a significant influence on the availability of aesthetic services, with Dingbian County in the west showing a notably high supply of aesthetic value. Educational services were predominantly at low (low and very low) supply levels, accounting for 95% of the areas assessed, whereas areas with high supply represented only 2%. Both recreational and spiritual services had similar supply proportions across different levels. High supply areas for recreational services were more concentrated in Yuyang District, while high supply areas for spiritual services were relatively dispersed (Figure 4).

Linear factors significantly affected the supply of CESs. Distance from the road was the most influential factor affecting CES supply capacity, making a cumulative contribution of 126.7%. Services such as recreational, spiritual, and educational services, which support human leisure activities, emotional needs, and value dependence, were all significantly influenced by road factors, highlighting the importance of road accessibility to CES spatial

distribution. Aesthetics reflect the natural landscape of the ecosystem, which was significantly influenced by the distance from the water. Moreover, land use accounted for 98.3% of the cumulative influence. In contrast to services characterized by ecological resource concentration, CESs, due to their cultural attributes, were more susceptible to human activities. The environmental factor with the lowest contribution rate was the distance from town, accounting for only 11%, indicating that the proximity to administrative centers had a minimal impact on the spatial distribution of CES supply (Table 3).

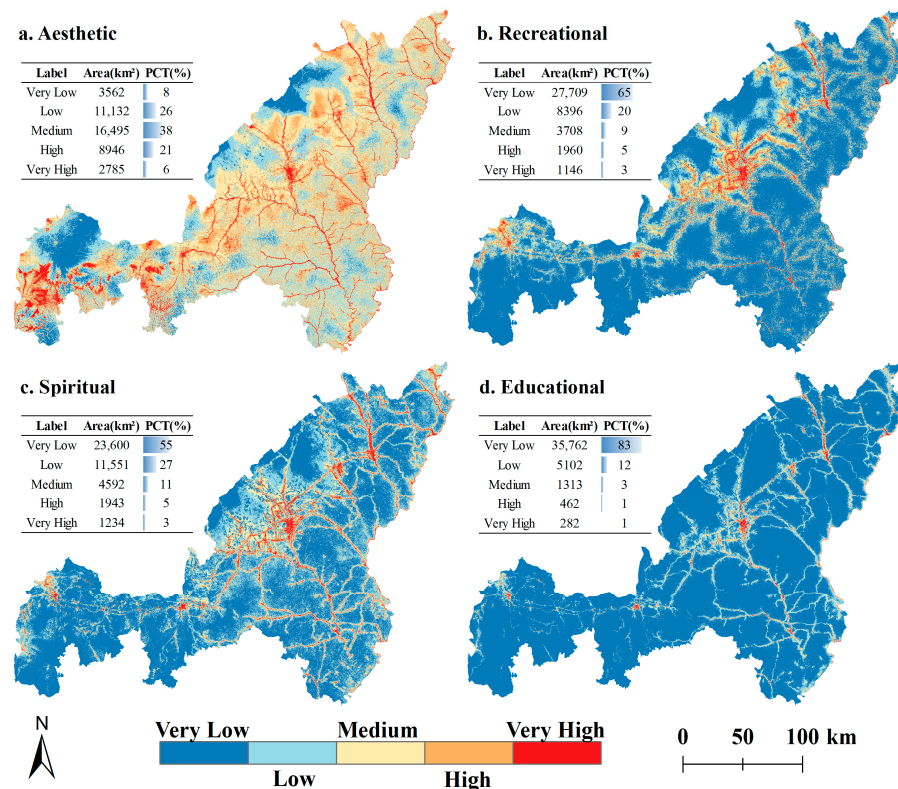


Figure 4. Spatial distribution of CES supply in Yulin City.

Table 3. Contribution of environmental factors (%).

CES	Environmental Factors						
	DisR	LU	DisW	NDVI	DEM	Slope	DisT
Aesthetic	7.8	17	38.3	3.8	23.9	6.8	2.3
Recreation	34	26.8	1.3	23.5	5.4	6.1	2.9
Spiritual	45.3	9.3	13.9	23.2	2.5	1.1	4.7
Educational	39.6	45.2	2	9.8	2	0.3	1.1
Cumulative contribution	126.7	98.3	55.5	60.3	33.8	14.3	11

DisR: distance from road; LU: land use; DisW: distance from water; NDVI: normalized difference vegetation index; DEM: digital elevation model; Slope: terrain slope; DisT: distance from town.

3.1.2. Characteristics of CES Demand

The 12 topics analyzed by the LDA model were visualized as word clouds, where the size of the words represented the frequency of its appearance in each topic (Figure 5).

Aesthetic and recreational services each encompassed four topics. The topic words for aesthetics focused on adjectives or nouns describing the Danxia landform and Wave Valley, showcasing the distinctive landscape features of northern Shaanxi. Recreational services primarily included content related to tourist experiences, photography, check-ins, and activities. Spiritual and educational services both contain two topics. The spiritual topic

included words associated with heritage, ancient cities, and other historical and cultural themes. The topic words for educational services were represented by children, parks, and museums, reflecting the region’s ecological and cultural education function (Table 4).

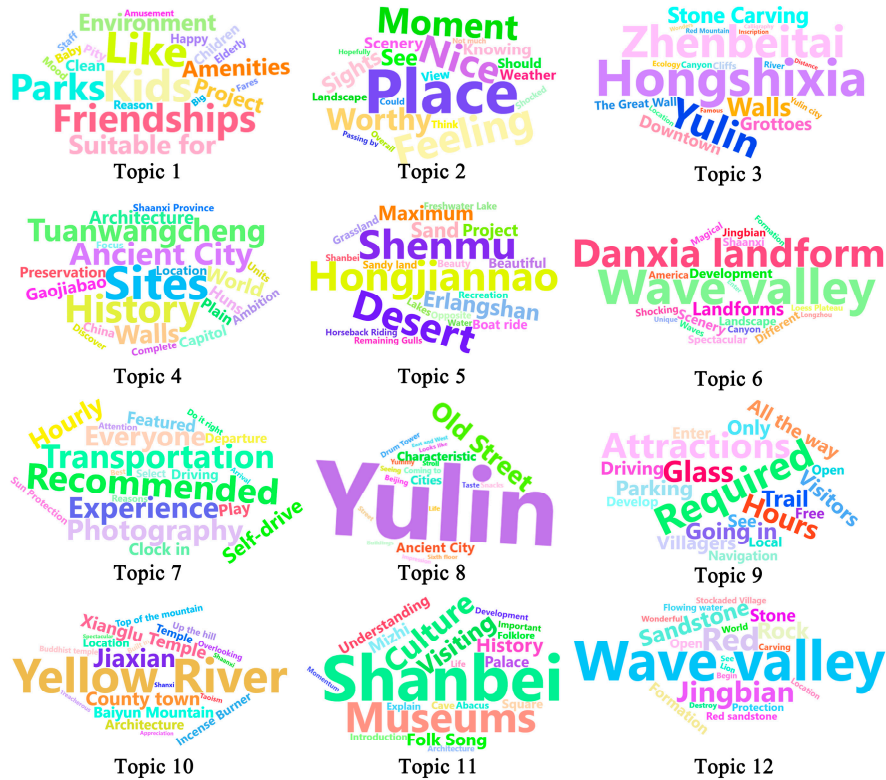


Figure 5. The word cloud of Topic 1~12.

Table 4. CES topics and words.

CES Types	Topics	Words
Aesthetic	Topic3, Topic5, Topic6, Topic12	Hongshixia, Zhenbeitai, Hongjiannao, Shenmu, Wave valley, Danxia landform, Red
Recreational	Topic2, Topic7, Topic8, Topic9	Place, Feeling, Recommended, Transportation, Yulin, Old Street, Required, Attractions
Spiritual	Topic4, Topic10	Sites, History, Ancient City, Yellow River, Jiaxian
Education	Topic1, Topic11	Kids, Like, Friendships, Shanbei, Museums, Culture

CES demand showed strong spatial clustering, with areas of high demand concentrated in Yuyang District and Jingbian County (Figure 6). Yuyang District, located at the center of Yulin City, functioned as a hub for the aggregation of spatial CES demand. Jingbian County, known as a major tourist destination in Shaanxi Province, featured natural landscapes such as Wave Valley, Danxia landforms, and the Mu Us Desert, alongside cultural attractions including revolutionary sites. Its numerous tourist attractions established it as a secondary hub for CES demand in Yulin City. The primary types of CES perceived in Yulin were spiritual services, which reflect local history and emotional attachment. Upon comparing the spatial distribution maps of CES demand, it was observed that spiritual services showed the strongest spatial connectivity. The areas with high demand for these services constituted the largest proportion at 15%, while the areas with low demand represented the smallest proportion at 65% (Figure 6d). Recreational services had the second largest proportion of high demand at 11%, and educational services had the lowest proportion of

high demand at only 6% (Figure 6b,c). Although there were more topics related to aesthetic services, these areas tended to be clustered, leading to an overall lower perception. The areas with low demand for aesthetics comprised the largest proportion at 86% (Figure 6a).

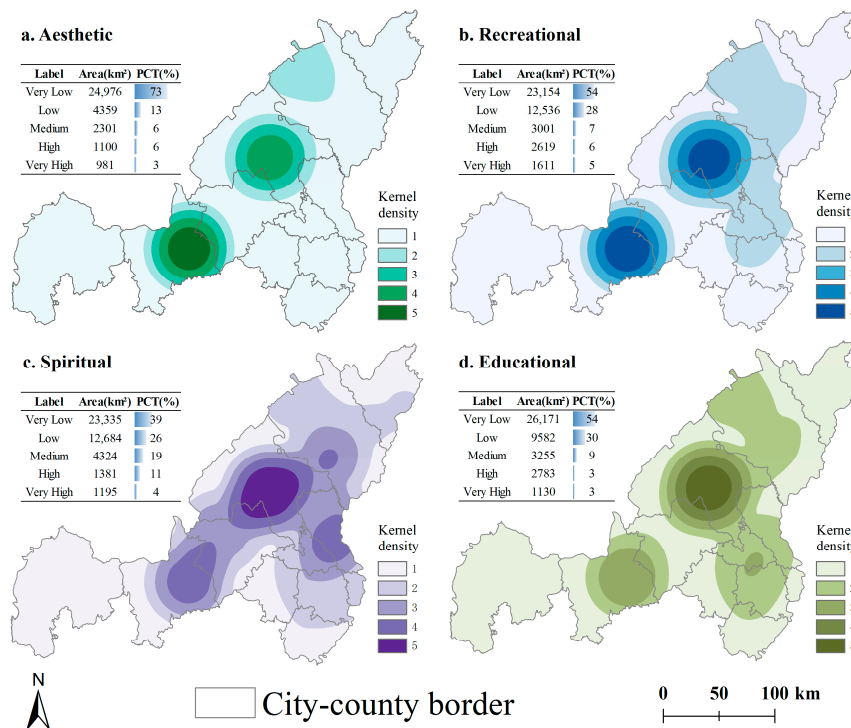


Figure 6. Spatial distribution of CES demand in Yulin City.

3.2. Mismatch of CES Supply–Demand

The distribution pattern of the ESDR was shown in Figure 7. A notable spatial imbalance was evident in the regional ESDR. Deficit areas were primarily located in population centers, such as Yuyang and Jingbian County, which possessed abundant basic resources and dense populations, exhibiting a high demand for CES. This resulted in the region exhibiting a characteristic where demand surpassed supply. Counties in the western and northern regions, characterized by sparse populations but abundant vegetation, displayed an excess of supply over demand. Therefore, it was crucial to attract tourists to areas with high supply and low demand through the development of tourism resources and infrastructure. The average ESDR values for aesthetic, recreational, spiritual, and educational services were 0.369, -0.038 , -0.056 , and -0.096 , respectively. The aesthetic service was able to match the regional demand well and generate a positive surplus. However, the ESDRs for the other services exhibited negative characteristics, suggesting that stakeholder demand for these services was insufficiently fulfilled. Recreational and spiritual services demonstrated a better alignment between supply and demand, as indicated by an average ESDR closer to 0 (Figure 7b). While the region exhibited a strong supply capacity for ecological landscapes, the most supply–demand mismatch was observed in aesthetic services, as reflected by the largest absolute value of the average ESDR (Figure 7a).

The cold- and hotspot areas of the ESDR displayed significant spatial clustering. All types of CESs followed the pattern of “cold spots at the center, hot spots at the periphery, with no distinct spatial clustering in the interior (Figure 8)”. The cold- and hotspots of the ESDR for spiritual services were the most pronounced, with 49 coldspot townships and 54 hotspot townships (Figure 8c). In contrast, the spatial concentration of aesthetic services was the least noticeable, with 19 coldspot townships and 20 hotspot townships (Figure 8a). This explained why the supply and demand for aesthetic services was the most poorly

matched. Due to the dual influence of demand agglomeration and supply fragmentation, the aesthetic services supply and demand failed to align effectively. There is a need to offer more options for CES supply–demand matching from the perspective of service flow.

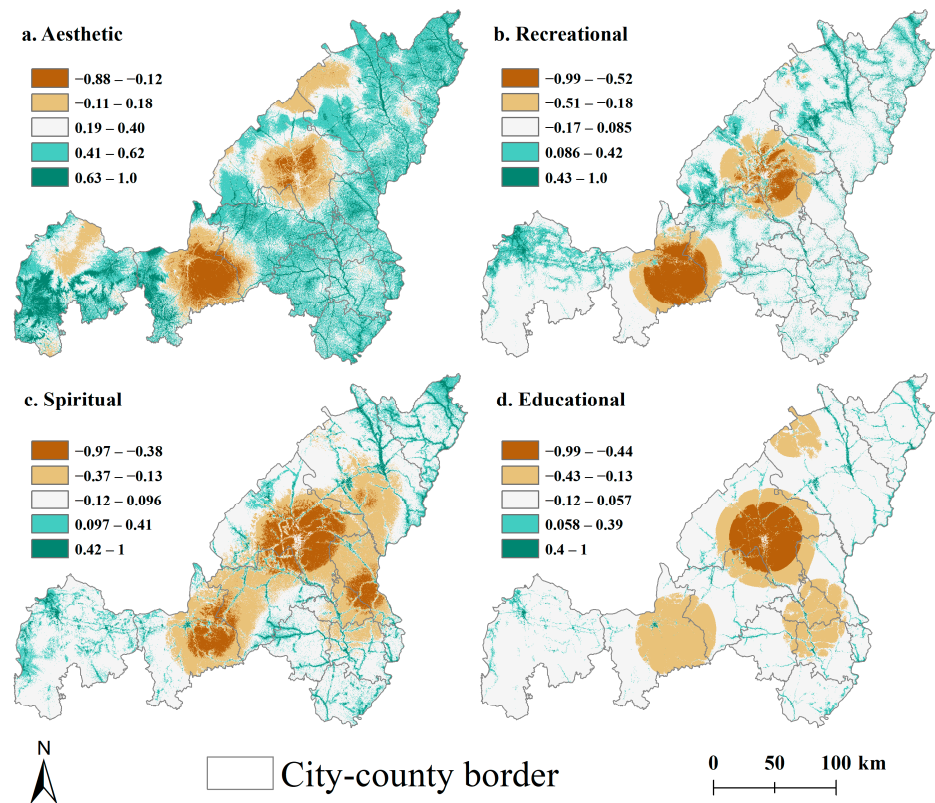


Figure 7. Spatial distribution of ESDR in Yulin City.

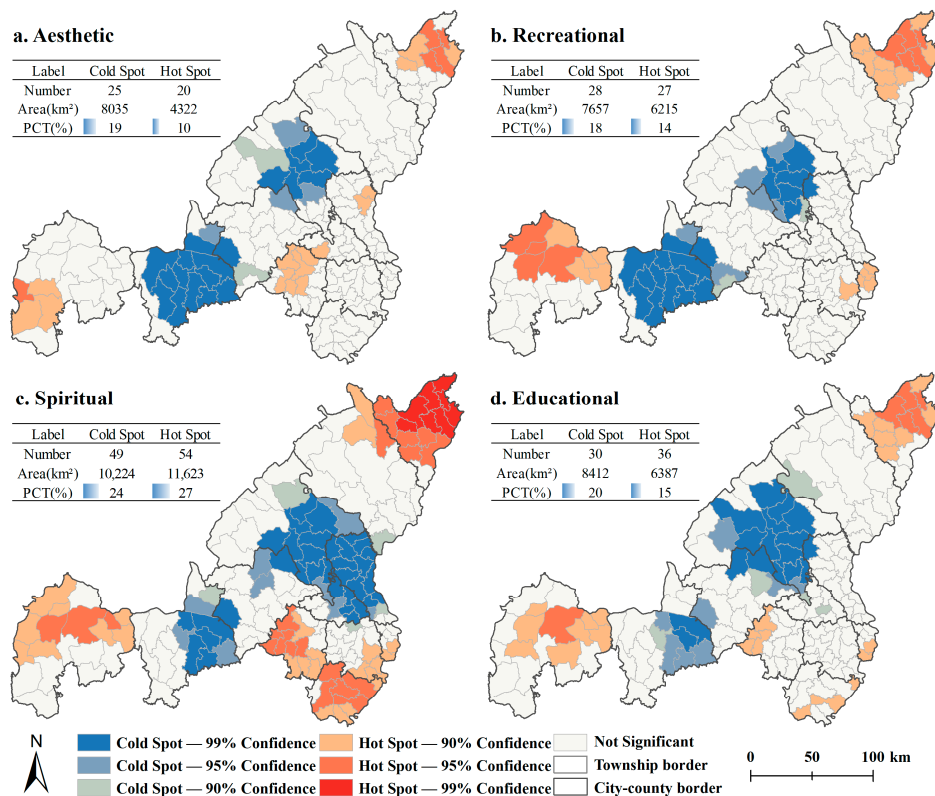


Figure 8. Cold and hotspots of ESDR.

3.3. Spatial Flow Characteristics of CES

The transmission range of each type of CES ranged from 10.87 to 254.32 km. The average flow intensity followed this order: aesthetic > spiritual > educational > recreational (Table 5). Aesthetic services, being the least matched, required a stronger flow to achieve spatial alignment. The surplus area was more capable of radiating towards the deficit area, and since the surplus area was situated on the fringe region, a significant portion of surplus value was transferred to regions beyond the confines of the study area (Figure A4).

Table 5. Parameters of CES flow.

Types	Transfer Count	Minimum Flow Radius (km)	Maximum Flow Radius (km)	Average Flow Intensity
Aesthetic	500	12.80	162.85	5.15
Recreational	756	11.75	254.32	0.21
Spiritual	2646	11.26	211.20	1.43
Educational	900	10.87	223.53	0.29

Figure 9 displayed the transfer lines of the top 200 CES spatial flows. It could be observed that, although the final beneficiary districts were concentrated in Yuyang and Jingbian, clear differences existed in the spatial mobility of the diverse types of CESs. Aesthetic service complement areas were situated near the deficit areas, while recreation services required a longer distance for the movement of services to achieve supply–demand matching (Figure 9a,b). The spiritual and educational deficit areas were recharged from the Zizhou and Wubao counties in the south, with the strong spatial flow for both services following a similar orientation, flowing from Macha Township in Zizhou and Zhangjiayan Township in Wubao to the central townships of Yuyang and Jingbian (Figure 9c,d).

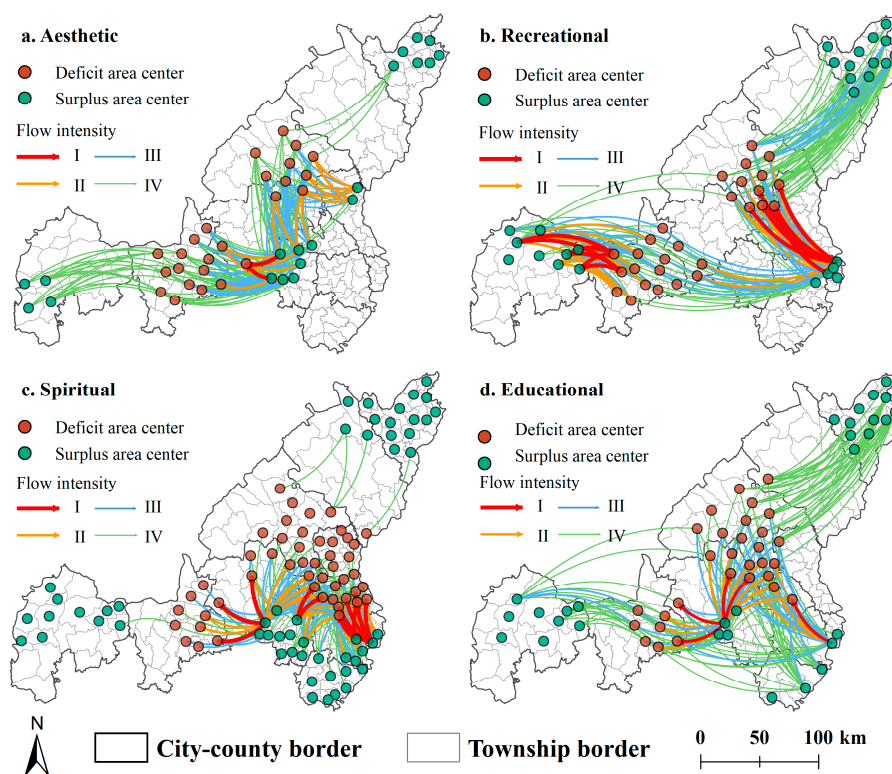


Figure 9. Direction and intensity of CES flow.

The transfer in and out of CES was characterized by a clear “gravitational pull”. Because of spatial distance limitations, areas with high transfer in and out were near each other, displaying a relative attraction (Figure 10). Among them, the highest output areas for aesthetic and spiritual services were both located in the townships of Zizhou County (Figure 10a,c), the highest output area for recreational services was in Chashang Township (Figure 10b), and the highest output area for educational services was in Zhangjiayan Township (Figure 10d). Even though the identifying scale of CES mobility was the township, a comparison of the spatial pattern of CES transfers revealed that CES transfers in and out almost did not coexist at the county level.

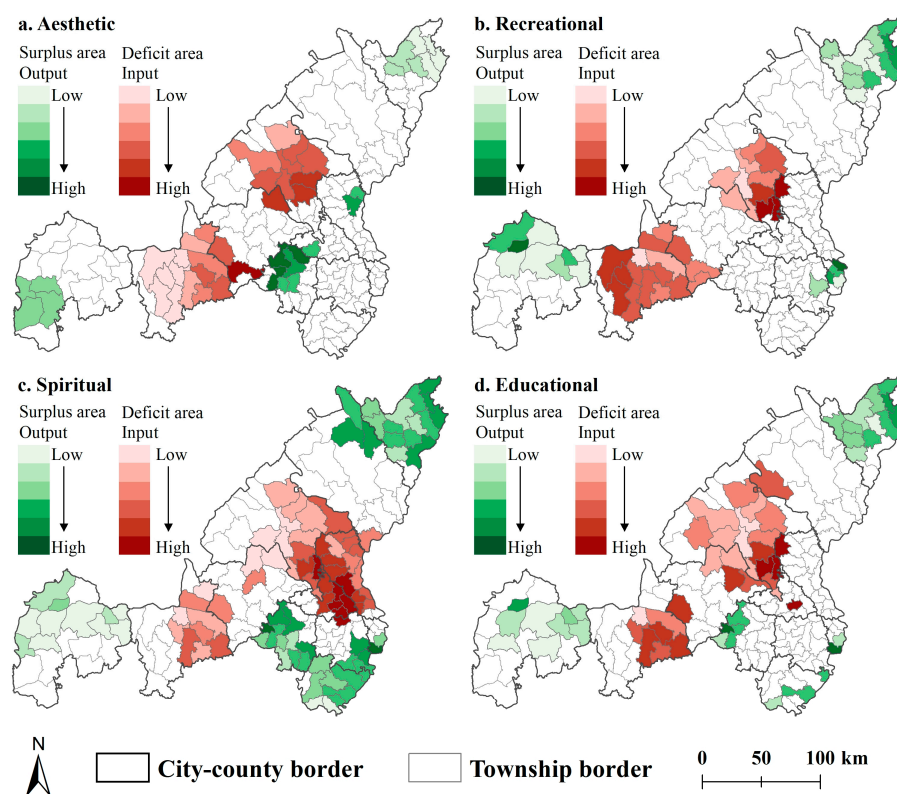


Figure 10. Transfer intensity of CESs.

The economic development level of counties affects CES supply–demand flow (Figure 11). Counties with lower economic levels, such as Suide, Wubu, Zizhou, and Qingjian, exhibited more surplus characteristics, whereas counties with higher economic levels, such as Hengshan, Jingbian, and Yuyang, displayed more deficit characteristics. The impact of distance from the city center should not be overlooked (Figure 11). Mizhi and Jiaxian, despite their lower economic levels, exhibited strong CES demand due to their proximity to the city center and the influence of urban radiation. In contrast, Dingbian and Fugu benefited from the ecological advantages of peripheral areas, maintaining economic growth while sustaining a surplus supply. These results remind us to focus on the development gaps among different counties, while considering the impact and limitations of distance, to achieve targeted guidance for CES flow from the demand side.

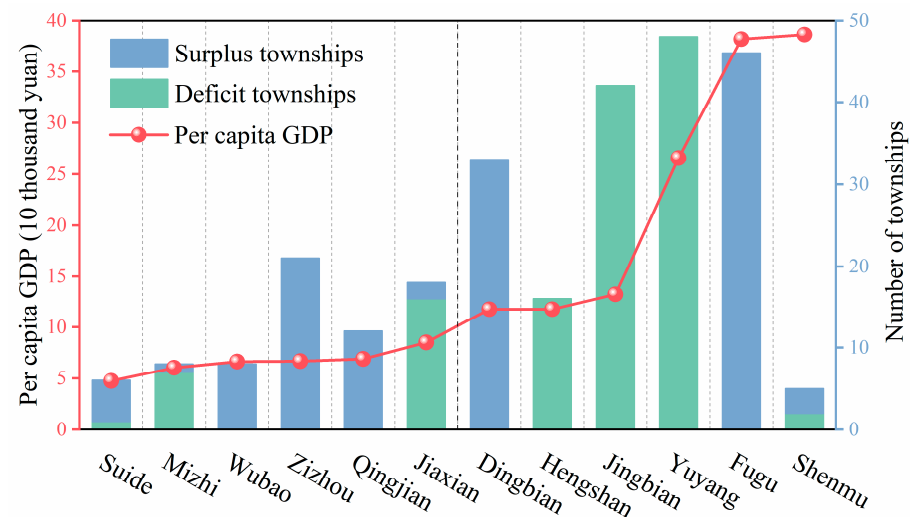


Figure 11. Per capita GDP and total number of surplus–deficit townships.

4. Discussion

4.1. CES Supply–Demand–Flow Characteristics

The supply–demand differences in various CESs are objectively determined by the inherent capabilities of ecosystems, while subjectively influenced by variations in public perception and preferences [76]. Yulin City is situated at the junction of the sandy grassland area and the Loess Plateau, characterized by its complex and diverse ecological landscapes. This unique setting provides a superior capacity for aesthetic services provision, creating conditions for future exploration of regional ecological value through eco-tourism. Moreover, beneficiaries of CESs exhibited a high demand for spiritual services, shaped by the intertwined influences of Yellow River civilization, Loess culture, and red culture. There is a need to emphasize the preservation and development of intangible cultural heritage, including folk songs, customs, heritage, and religious practices, as well as paying attention to services that address the intangible dimensions of the human–nature relationship.

Similar to the global trend of high demand and low supply observed in urban centers of metropolitan areas [77], tourist zones [14], and economically developed regions [78], the spatial distribution of CESs in the study area was typified by significantly more demand than supply in central areas and more supply than demand in remote areas. The concentration of population and resources in the city center generated strong demand, and the distribution of high-density buildings further encroached upon ecological space and weakened the supply of CESs. Counties located farther from the city center were less affected by economic radiation, and their delayed development led to a diminished awareness of CESs [79]. Additionally, the ecological advantages stemming from the extensive ecological land area and dense vegetation cover in these regions significantly bolstered the supply of CESs. Consequently, CES flow predominantly occurred from remote to central regions and from less developed to more developed areas. Economically advanced core counties attracted a greater share of CES flow [80]. Recognizing CES flow contributes to balancing supply and demand, thereby promoting a healthy and equitable ecosystem [79].

It is crucial to acknowledge that trade-offs and synergies frequently occur both within CESs and between CESs and other ecosystem services [3,81]. This demonstrates the connectivity and integrity within the ecosystem [82]. A notable spatial overlap was observed between various CES supply–demand patterns, influenced by the concentration of infrastructure in city centers. This overlap subsequently resulted in similarities in the direction of CES spatial flow. Focusing on CES trade-off synergies allows management and policy-

makers to adopt a broader perspective and develop policies that transcend the limitations of a single CES [83,84].

4.2. Recommendations for Urban Landscapes Management

4.2.1. Enhancing the Functionality of Low-Perception CESs

Whether in fast-urbanizing areas or ecologically fragile zones, aesthetic services were repeatedly recognized as a highly perceptible service category [10,33]. In our study, aesthetic services demonstrated both a high supply level and a large proportion of demand topics. Aesthetic services represent the material attributes of CESs, encompassing elements such as mountains, rivers, and scenic spots, which intuitively reflect the interactions between humankind and natural world. This makes them a direct manifestation of the ecological landscape's value, rendering them the most recognizable and perceptible aspects. In contrast, educational services had lower perceptual capabilities, indicating that the ecological value related to science education had not been fully met, thus limiting the comprehensive development of CESs. More nature-based science education initiatives should be implemented to provide residents greater opportunities to understand and respect the ecosystem, fostering interest in scientific research and conservation [85].

4.2.2. Alleviating the Supply–Demand Imbalance of CES

Compared to provincial or central cities [78,86], small cities located in remote areas with underdeveloped economies are more susceptible to the “siphon effect” of city centers, where excessive concentration of infrastructure and population results in high demand for CESs. Adopting localized approaches to address the supply–demand mismatch in such cities is beneficial for achieving landscape sustainability [87].

Inadequate transportation infrastructure has impacted the public's ability to access CESs, leading to areas with high supply but low demand. To address this issue, improving transportation accessibility and intensifying promotional efforts to enhance the visibility of scenic spots would help attract more visitors. For example, Yulin and Dingbian County possessed exceptional ecological characteristics that needed adequate promotion and development due to natural constraints. To fully utilize its potential, it is important to explore local characteristics and integrate traditional culture into the ecological space, thereby enhancing the region's attractiveness. However, specific strategies, such as enhancing accessibility and infrastructure, may shift the region from a high supply and low demand scenario to one characterized by low supply and high demand. Therefore, further exploration is necessary to identify the critical threshold for such transitions [88]. City centers faced insufficient CES facilities due to constraints on available construction land. Local ecological restoration projects are implemented based on natural conditions. Roof greening and three-dimensional greening are being developed appropriately to enhance the accessibility of street green spaces. These efforts also aim to increase the clustering function and scale effects of CESs in the central district.

4.2.3. Optimizing the Spatial Flow of CESs

The supply–demand mismatch was caused by the disparity among the availability of resources and the practical ability to access those resources [89]. The service value of CESs can only be realized when their flow spatially reaches deficit areas. Due to the gradient effect of urban and rural development, counties at different stages of development focus on CES supply and demand in distinct ways. It is imperative to address the disparities between counties, balance the geographical correlation between surplus and deficit areas, clarify the scope and intensity of CES spatial transfer, and differentiate and guide the spatial flow of CESs. The complex and dynamic changes inherent in urban ecosystems necessitate the integration of the concept of flow feedback into the development of ecological frameworks,

along with the sharing of CES resources through spatial flows across regions. At the same time, it encourages the enhancement of the multifaceted transit network linking the central area with surrounding regions as well as the inner-city and outer-city areas and reduces the impact of distance-related costs, thereby expanding the transmission capacity of CESs, enhancing the efficiency of the use of surplus areas and strengthening the function of urban CESs.

4.3. Limitations and Uncertainties

The supply–demand balance of ESs is fundamental to urban sustainability. Using multi-source data coupled with multi-model approaches to deduce human values and experiences pertaining to the environment is a worthwhile method [90]. Firstly, we applied user interaction data from social media throughout the entire process of CES supply and demand, which constitutes the main contribution of this study. Previous research primarily identified supply points for CESs through surveys, while the identification of demand was derived from social media texts or images [86,90]. These studies required substantial time and manpower during the initial survey phase and overlooked the multifaceted nature of CESs. In our study, we introduced POI data to assess CES supply, with both supply and demand layers derived from social media data sources. This analysis within the same dimension reduces the biases associated with traditional survey and saves costs related to questionnaire collection. Additionally, user interaction data provide a wealth of collective information, enriching our comprehension of diverse CES supply and demand, addressing the previous neglect of cultural services, such as spiritual and educational services [46]. User interaction data have gradually emerged as a new data source for CES research [91]. Secondly, the identification of CES supply–demand–flow characteristics relied on the effective outputs of multi-source models. The Maxent model provides reliable spatial outputs that reveal the extent to which various environmental factors influence cultural service supply. The LDA model integrates vast textual resources into the evaluation model, which reflect beneficiaries’ attitudes and preferences towards CESs. It establishes a three-tier hypothesis of document–topic–word, determines the probability of different topics of a document, and realizes the recognition of multiple CES types [27,92]. The breakpoint model and gravity model effectively explained the range, intensity, and magnitude of CES spatial flow, revealing the distinctions and relationships in the spatial transfer of different CESs. Finally, the integrated approach of coupling multi-source models transcended the limitations of traditional ecological zones, parks, and urban green spaces, enabling a comprehensive assessment and facilitating the assessment and analysis of CESs across multiple scales [27,93]. It provided valuable information for regional landscape planning and urban development.

However, the research methods also have some limitations. For instance, the composition of social media users may be biased, as older individuals and children who do not frequently use the internet might be overlooked [94]. The LDA model’s text classification and the standards for classifying CESs may differ due to uncertainties in CES perception [78]. Participatory research conducted at the user level can be utilized to mitigate the biases caused by social media data. It is essential to enhance public understanding and awareness of ecosystems to alleviate uncertainties in CES perception. Additionally, the spatial conversion coefficient in the gravity model was derived from previous research, without fully considering the conceivable effects of environmental or societal elements on CES flow. In the future, there is a need to construct a scientifically comprehensible and scalable model to conduct a more refined assessment. The flow of resources from natural ecosystems to human social systems constitutes the supply and demand of CESs, and changes in resource and population dynamics over time may result in swift responses

in CES supply and demand matching. Future research on CESs should focus on ecosystem service management, combining qualitative survey data with social media data, and exploring the generation, flow, and benefits of CES from a temporal perspective [93].

5. Conclusions

In the context of growing cultural demands, CESs have played an increasingly significant role in daily life. How to effectively quantify CESs for ecological resource management requires sufficient attention. Our study applied geospatial data and user interaction data, coupled with multi-source models, to achieve the spatial quantification of CES supply–demand flow, providing a paradigm for CES research across various scales and regions. The results indicated the following: (1) there were perceptual differences across CESs, and aesthetic services was the most strongly perceived type of CES in Yulin; (2) there was a mismatch between supply and demand in Yulin, with a supply–demand deficit in the central area and a supply–demand surplus in the peripheral areas; (3) the replenishment from the surplus urban periphery to the deficit central urban area was the main characteristic of CES flow, with the flow of aesthetic and spiritual services being more active in Yulin. The combination of spatial–temporal big data and semantic analysis enabled the extraction of information regarding the recognition and utilization of ecological infrastructures from a CES perspective, contributing to the cultural benefits of people’s interaction with urban ecosystems. In the future, attention should be directed to the deeper causes of the coupled interaction between humanity and nature, analyzing the influencing factors of CES supply–demand patterns from the perspective of causality and flow response, and providing diverse references for effective urban landscape management.

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Data Availability Statement: The original contributions presented in this study are included in this article, further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

We selected seven environmental factors (Figure A1). There is a strong link between land use and ecosystems, and DEM and slope have obvious effects on the spatial distribution of CESs. Distance factors reflect the spatial relationship between CESs and resources, and vegetation cover responds to the hidden characteristics of space.

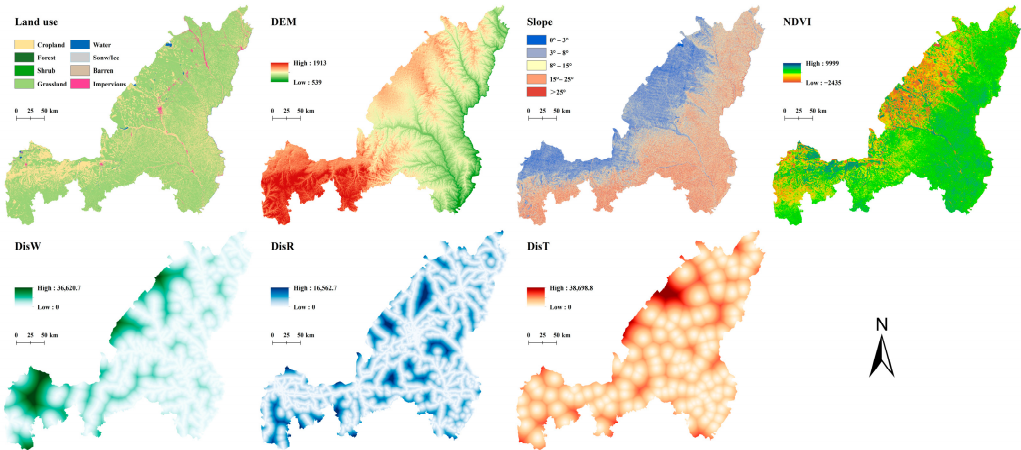


Figure A1. Environmental factors.

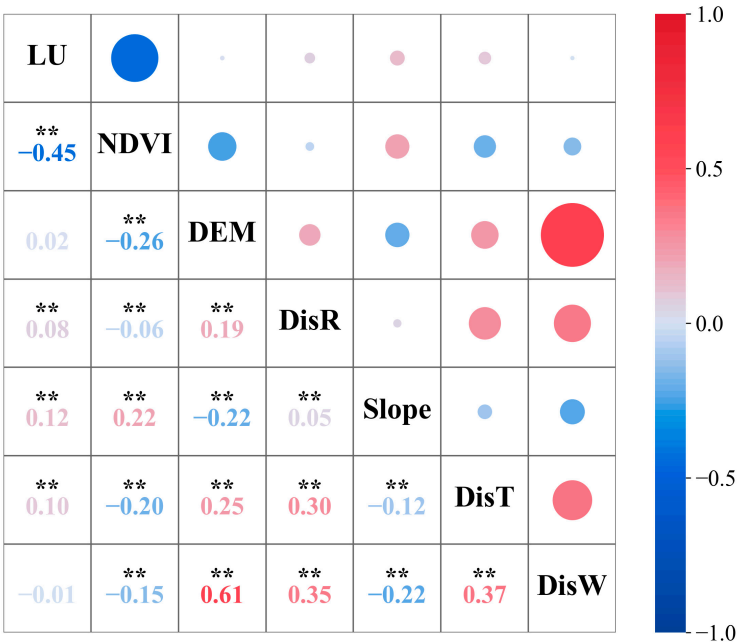


Figure A2. Correlation analysis of environmental factors (** $p < 0.01$). We used the Spearman correlation coefficient for the analysis. The results showed that the correlation coefficients for all environmental factors were less than 0.8, indicating that all environmental factors passed the correlation test.

Perplexity is an important metric for evaluating topic models such as LDA. It reflects the model’s ability to explain the data; the lower the perplexity, the better the model’s ability to explain the data. The formula is as follows:

$$Perplexity(D) = \exp\left\{-\left(\sum_{d=1}^M \log P(w_d) / \sum_{d=1}^M N_d\right)\right\} \tag{A1}$$

where D is the test set in the corpus; M is the number of documents; N_d is the total number of words in the entire documents; w_d is the word in document d ; $P(w_d)$ is the generation probability of the w_d in the document d . We used the sklearn toolkit in python 3.12 to calculate the perplexity. The results indicated that the perplexity was lowest when the number of topics was 12 (Figure A3).

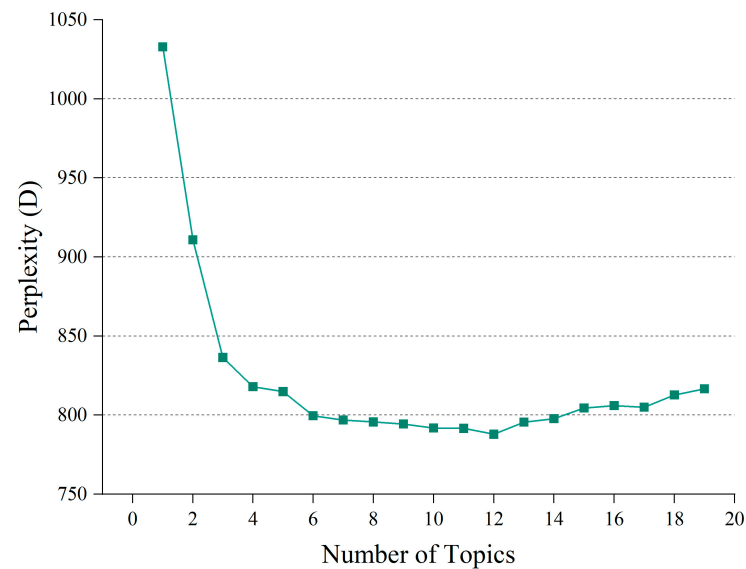


Figure A3. Perplexity curve.

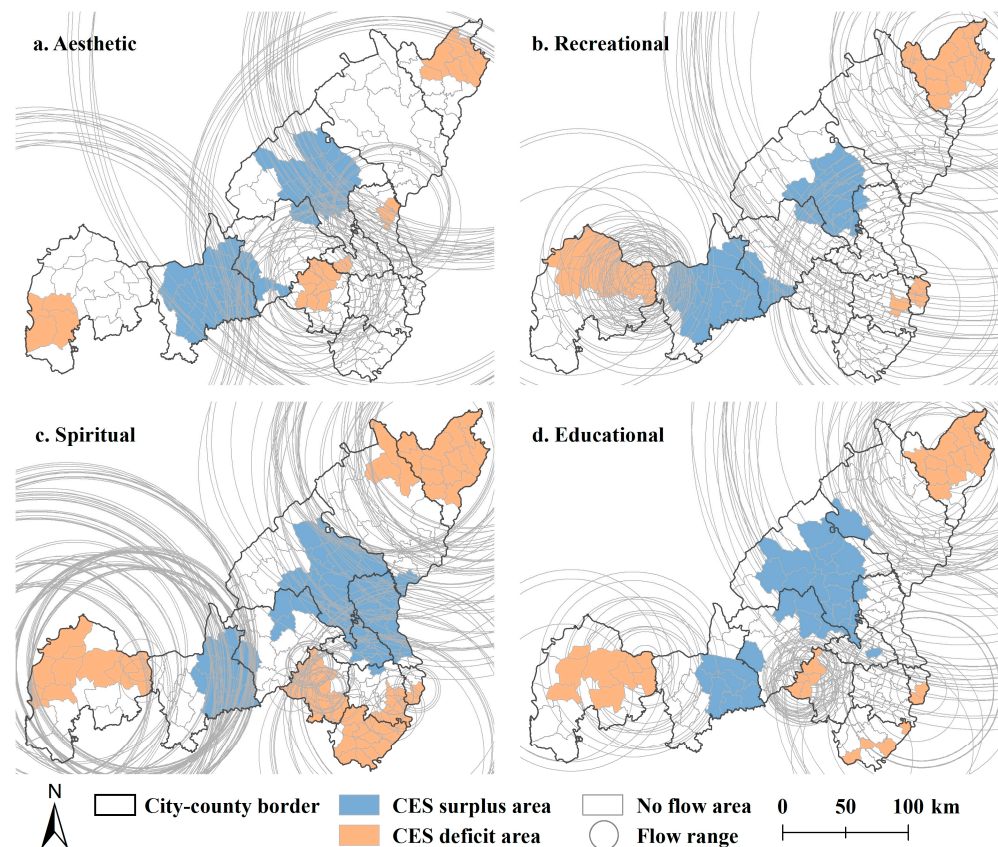


Figure A4. Range of CES.

References

1. MEA. *Millennium Ecosystem Assessment: Ecosystems and Human Well-Being*; Island Press: Washington, DC, USA, 2005.
2. Kosanic, A.; Petzold, J. A systematic review of cultural ecosystem services and human wellbeing. *Ecosyst. Serv.* **2020**, *45*, 101168. [\[CrossRef\]](#)
3. Daniel, T.C.; Muhar, A.; Arnberger, A.; Aznar, O.; Boyd, J.W.; Chan, K.M.A.; Costanza, R.; Elmquist, T.; Flint, C.G.; Gobster, P.H.; et al. Contributions of cultural services to the ecosystem services agenda. *Proc. Natl. Acad. Sci. USA* **2012**, *109*, 8812–8819. [\[CrossRef\]](#)

4. Margaryan, L.; Prince, S.; Ioannides, D.; Röslmaier, M. Dancing with cranes: A humanist perspective on cultural ecosystem services of wetlands. *Tour. Geogr.* **2018**, *24*, 501–522. [\[CrossRef\]](#)
5. Wood, S.L.R.; Jones, S.K.; Johnson, J.A.; Brauman, K.A.; Chaplin-Kramer, R.; Fremier, A.; Girvetz, E.; Gordon, L.J.; Kappel, C.V.; Mandle, L.; et al. Distilling the role of ecosystem services in the Sustainable Development Goals. *Ecosyst. Serv.* **2018**, *29*, 70–82. [\[CrossRef\]](#)
6. Pan, Y.; Qu, Y. Cultural Ecosystem Services in Land Use/Land Cover Change: A Literature Review and Prospects for Future Research. *Land* **2024**, *13*, 2027. [\[CrossRef\]](#)
7. Zhao, Q.; Li, J.; Liu, J.; Cuan, Y.; Zhang, C. Integrating supply and demand in cultural ecosystem services assessment: A case study of Cuihua Mountain (China). *Environ. Sci. Pollut. Res. Int.* **2019**, *26*, 6065–6076. [\[CrossRef\]](#)
8. Wang, L.; Zheng, H.; Wen, Z.; Liu, L.; Robinson, B.E.; Li, R.; Li, C.; Kong, L. Ecosystem service synergies/trade-offs informing the supply-demand match of ecosystem services: Framework and application. *Ecosyst. Serv.* **2019**, *37*, 100939. [\[CrossRef\]](#)
9. Caputo, J.; Butler, B. Ecosystem service supply and capacity on U.S. family forestlands. *Forests* **2017**, *8*, 395. [\[CrossRef\]](#)
10. Meng, S.T.; Huang, Q.X.; Zhang, L.; He, C.Y.; Inostroza, L.; Bai, Y.S.; Yin, D. Matches and mismatches between the supply of and demand for cultural ecosystem services in rapidly urbanizing watersheds: A case study in the Guanting Reservoir basin, China. *Ecosyst. Serv.* **2020**, *45*, 101156. [\[CrossRef\]](#)
11. Jiang, W. Ecosystem services research in China: A critical review. *Ecosyst. Serv.* **2017**, *26*, 10–16. [\[CrossRef\]](#)
12. Wei, H.; Liu, H.; Xu, Z.; Ren, J.; Lu, N.; Fan, W.; Zhang, P.; Dong, X. Linking ecosystem services supply, social demand and human well-being in a typical mountain–oasis–desert area, Xinjiang, China. *Ecosyst. Serv.* **2018**, *31*, 44–57. [\[CrossRef\]](#)
13. Li, J.; Li, K.; Wang, Y.; Jiao, R. Comparative Study on the Perception of Cultural Ecosystem Services in Taibai Mountain National Forest Park from Different Stakeholder Perspectives. *Land* **2024**, *13*, 2207. [\[CrossRef\]](#)
14. Lorilla, R.S.; Kalogirou, S.; Poirazidis, K.; Kefalas, G. Identifying spatial mismatches between the supply and demand of ecosystem services to achieve a sustainable management regime in the Ionian Islands (Western Greece). *Land Use Pol.* **2019**, *88*, 104171. [\[CrossRef\]](#)
15. Lou, C.; Wang, F.; Li, G.; Yang, C. Identifying supply-demand spatial matches of cultural ecosystem services provided by urban parks in polycentric mountainous cities. *Acta Ecol. Sin.* **2024**, *44*, 5816–5827. [\[CrossRef\]](#)
16. Hernández-Morcillo, M.; Plieninger, T.; Bieling, C. An empirical review of cultural ecosystem service indicators. *Ecol. Indic.* **2013**, *29*, 434–444. [\[CrossRef\]](#)
17. Wartmann, F.M.; Purves, R.S. Investigating sense of place as a cultural ecosystem service in different landscapes through the lens of language. *Landsc. Urban Plan.* **2018**, *175*, 169–183. [\[CrossRef\]](#)
18. Riechers, M.; Barkmann, J.; Tschardt, T. Diverging perceptions by social groups on cultural ecosystem services provided by urban green. *Landsc. Urban Plan.* **2018**, *175*, 161–168. [\[CrossRef\]](#)
19. Balázs, Á.; Dänhardt, J.; Collins, S.; Schweiger, O.; Settele, J.; Hartel, T. Understanding cultural ecosystem services related to farmlands: Expert survey in Europe. *Land Use Pol.* **2021**, *100*, 104900. [\[CrossRef\]](#)
20. Jeuring, J.H.G. Weather perceptions, holiday satisfaction and perceived attractiveness of domestic vacationing in The Netherlands. *Tour. Manag.* **2017**, *61*, 70–81. [\[CrossRef\]](#)
21. Zhao, J.; Zhang, N.; Liu, Z.; Zhang, Q.; Shang, C. SWAT model applications: From hydrological processes to ecosystem services. *Sci. Total Environ.* **2024**, *931*, 172605. [\[CrossRef\]](#)
22. Lin, Y.; Dong, Z.; Zhang, W.; Zhang, H. Estimating inter-regional payments for ecosystem services: Taking China's Beijing-Tianjin-Hebei region as an example. *Ecol. Econ.* **2020**, *168*, 106514. [\[CrossRef\]](#)
23. Johnson, G.W.; Bagstad, K.J.; Snapp, R.R.; Villa, F. Service Path Attribution Networks (SPANs). *Int. J. Agric. Environ.* **2012**, *3*, 54–71. [\[CrossRef\]](#)
24. Ouyang, X.; Wang, Z.; Zhu, X. Construction of the Ecological Security Pattern of Urban Agglomeration under the Framework of Supply and Demand of Ecosystem Services Using Bayesian Network Machine Learning: Case Study of the Changsha–Zhuzhou–Xiangtan Urban Agglomeration, China. *Sustainability* **2019**, *11*, 6416. [\[CrossRef\]](#)
25. Shi, T.; Gao, F. Utilizing Multi-Source Geospatial Big Data to Examine How Environmental Factors Attract Outdoor Jogging Activities. *Remote Sens.* **2024**, *16*, 3056. [\[CrossRef\]](#)
26. Richards, D.R.; Tunçer, B. Using image recognition to automate assessment of cultural ecosystem services from social media photographs. *Ecosyst. Serv.* **2018**, *31*, 318–325. [\[CrossRef\]](#)
27. Wang, L.; Huang, L.; Cao, W.; Zhai, J.; Fan, J. Assessing grassland cultural ecosystem services supply and demand for promoting the sustainable realization of grassland cultural values. *Sci. Total Environ.* **2024**, *912*, 169255. [\[CrossRef\]](#)
28. Yoshimura, N.; Hiura, T. Demand and supply of cultural ecosystem services: Use of geotagged photos to map the aesthetic value of landscapes in Hokkaido. *Ecosyst. Serv.* **2017**, *24*, 68–78. [\[CrossRef\]](#)
29. Sinclair, M.; Ghermandi, A.; Sheela, A.M. A crowdsourced valuation of recreational ecosystem services using social media data: An application to a tropical wetland in India. *Sci. Total Environ.* **2018**, *642*, 356–365. [\[CrossRef\]](#)

30. Inácio, M.; Gomes, E.; Bogdzevič, K.; Kalinauskas, M.; Zhao, W.; Pereira, P. Mapping and assessing coastal recreation cultural ecosystem services supply, flow, and demand in Lithuania. *J. Environ. Manag.* **2022**, *323*, 116175. [CrossRef]
31. Ghermandi, A.; Sinclair, M. Passive crowdsourcing of social media in environmental research: A systematic map. *Glob. Environ. Change* **2019**, *55*, 36–47. [CrossRef]
32. Zhao, Y.; You, W.; Lin, X.; He, D. Assessing the supply and demand linkage of cultural ecosystem services in a typical county-level city with protected areas in China. *Ecol. Indic.* **2023**, *147*, 109992. [CrossRef]
33. Zhang, J.; He, C.; Huang, Q.; Li, J.; Qi, T. Evaluating the supply and demand of cultural ecosystem services in the Tibetan Plateau of China. *Landsc. Ecol.* **2022**, *37*, 2131–2148. [CrossRef]
34. Sherrouse, B.C.; Semmens, D.J.; Ancona, Z.H. Social Values for Ecosystem Services (SolVES): Open-source spatial modeling of cultural services. *Environ. Modell. Softw.* **2022**, *148*, 105259. [CrossRef]
35. Shi, X.; Zhao, X.; Pu, J.; Gu, Z.; Zhao, Q.; Feng, Y.; Zhou, S. Creating a monetization-SolVES model to visualize the cultural ecosystem services for promoting landscape conservation. *J. Nat. Conserv.* **2024**, *77*, 126521. [CrossRef]
36. Yu, D.; Xiang, B. Discovering topics and trends in the field of Artificial Intelligence: Using LDA topic modeling. *Expert Syst. Appl.* **2023**, *225*, 120114. [CrossRef]
37. Sun, L.; Yin, Y. Discovering themes and trends in transportation research using topic modeling. *Transp. Res. Part C Emerg. Technol.* **2017**, *77*, 49–66. [CrossRef]
38. Zhang, J.; He, C.; Huang, Q.; Li, L. Understanding ecosystem service flows through the metacoupling framework. *Ecol. Indic.* **2023**, *151*, 110303. [CrossRef]
39. Du, H.; Zhao, L.; Zhang, P.; Li, J.; Yu, S. Ecological compensation in the Beijing-Tianjin-Hebei region based on ecosystem services flow. *J. Environ. Manag.* **2023**, *331*, 117230. [CrossRef]
40. Yang, H.; Sun, L.; Zhou, M.; Li, F. Trade-off analyses of ecosystem services during the reconstruction of grain production space in Loess Plateau: A case of Yulin City. *Arid Land Geogr.* **2022**, *45*, 226–236. [CrossRef]
41. National Forestry and Grassland Administration. Available online: <https://www.forestry.gov.cn/c/www/dfdt/545006.jhtml> (accessed on 10 December 2023).
42. Li, Y.; Cheng, W.; Zuo, W.; Zhang, L. Agricultural Vulnerability to Drought in China's Agro-pastoral Ecotone: A Case Study of Yulin City, Shaanxi Province. *Chin. Geogr. Sci.* **2023**, *33*, 934–945. [CrossRef]
43. Shaanxi Provincial Cultural Heritage Administration. Available online: <https://www.shaanxi.gov.cn/wbxx/bkydww/lswhmc/> (accessed on 10 December 2023).
44. Pan, J.; Ma, Y.; Cai, S.; Chen, Y.; Chen, Y. Distribution patterns of lake-wetland cultural ecosystem services in highland. *Environ. Dev.* **2022**, *44*, 100754. [CrossRef]
45. Hynes, S.; Ghermandi, A.; Norton, D.; Williams, H. Marine recreational ecosystem service value estimation: A meta-analysis with cultural considerations. *Ecosyst. Serv.* **2018**, *31*, 410–419. [CrossRef]
46. Kalinauskas, M.; Bogdzevic, K.; Gomes, E.; Inacio, M.; Barcelo, D.; Zhao, W.; Pereira, P. Mapping and assessment of recreational cultural ecosystem services supply and demand in Vilnius (Lithuania). *Sci. Total Environ.* **2023**, *855*, 158590. [CrossRef]
47. Khosravi Mashizi, A.; Sharafatmandrad, M.; Alizadeh, R. Exploring plant diversity and aesthetic service of mountain and plain landscapes in semiarid ecosystems. *J. Nat. Conserv.* **2024**, *82*, 126740. [CrossRef]
48. TEEB. *The Economics of Ecosystems and Biodiversity*; The United Nations Environment Programme: Nairobi, Kenya, 2010.
49. Web Services API Downloads. Available online: <https://lbs.amap.com/api/webservice/download> (accessed on 25 March 2025).
50. Li, Y.; Zhang, D.; Wang, K.; Huang, L. Supply-demand relationships of cultural ecosystem services in rural areas: A case study of Huzhou City, Zhejiang Province. *Acta Ecol. Sin.* **2022**, *42*, 6888–6899. [CrossRef]
51. Zhao, Z.; Xiao, N.; Shen, M.; Li, J. Comparison between optimized MaxEnt and random forest modeling in predicting potential distribution: A case study with *Quasipaa boulengeri* in China. *Sci. Total Environ.* **2022**, *842*, 156867. [CrossRef]
52. Zhang, K.; Yao, L.; Meng, J.; Tao, J. Maxent modeling for predicting the potential geographical distribution of two peony species under climate change. *Sci. Total Environ.* **2018**, *634*, 1326–1334. [CrossRef]
53. Ma, M.; Yuan, W.; Ge, Q.; Yuan, W.; Yang, L.; Li, H.; Li, M. Big data analysis of social development situation in regions along the Belt and Road. *Prog. Geogr.* **2019**, *38*, 1009–1020. [CrossRef]
54. Zhang, B.; Wang, H. A new type of dual-scale neighborhood based on vectorization for cellular automata models. *GISci. Remote Sens.* **2021**, *58*, 386–404. [CrossRef]
55. Li, Y.; Li, Z.; Wang, J.; Zeng, H. Analyses of driving factors on the spatial variations in regional eco-environmental quality using two types of species distribution models: A case study of Minjiang River Basin, China. *Ecol. Indic.* **2022**, *139*, 108980. [CrossRef]
56. Phillips, S.J.; Anderson, R.P.; Schapire, R.E. Maximum entropy modeling of species geographic distributions. *Ecol. Modell.* **2006**, *190*, 231–259. [CrossRef]
57. Fielding, A.H.; Bell, J.F. A review of methods for the assessment of prediction errors in conservation presence/absence models. *Environ. Conserv.* **2002**, *24*, 38–49. [CrossRef]

58. Phillips, S.J.; Anderson, R.P.; Dudík, M.; Schapire, R.E.; Blair, M.E. Opening the black box: An open-source release of Maxent. *Ecography* **2017**, *40*, 887–893. [\[CrossRef\]](#)
59. Kong, W.; Li, X.; Zou, H. Optimizing MaxEnt model in the prediction of species distribution. *Chin. J. Appl. Ecol.* **2019**, *30*, 2116–2128. [\[CrossRef\]](#)
60. Porfirio, L.L.; Harris, R.M.B.; Lefroy, E.C.; Hugh, S.; Gould, S.F.; Lee, G.; Bindoff, N.L.; Mackey, B. Improving the Use of Species Distribution Models in Conservation Planning and Management under Climate Change. *PLoS ONE* **2014**, *9*, e113749. [\[CrossRef\]](#)
61. Farkhod, A.; Abdusalomov, A.; Makhmudov, F.; Cho, Y.I. LDA-Based Topic Modeling Sentiment Analysis Using Topic/Document/Sentence (TDS) Model. *Appl. Sci.* **2021**, *11*, 11091. [\[CrossRef\]](#)
62. Elsayed, A.; Rixon, S.; Zeuner, C.; Levison, J.; Binns, A.; Goel, P. Text mining-aided meta-research on nutrient dynamics in surface water and groundwater: Popular topics and perceived gaps. *J. Hydrol.* **2023**, *626*, 130338. [\[CrossRef\]](#)
63. Li, J.; Jiang, H.; Bai, Y.; Alatalo, J.M.; Li, X.; Jiang, H.; Liu, G.; Xu, J. Indicators for spatial-temporal comparisons of ecosystem service status between regions: A case study of the Taihu River Basin, China. *Ecol. Indic.* **2016**, *60*, 1008–1016. [\[CrossRef\]](#)
64. Pan, Z.; He, J.; Liu, D.; Wang, J.; Guo, X. Ecosystem health assessment based on ecological integrity and ecosystem services demand in the Middle Reaches of the Yangtze River Economic Belt, China. *Sci. Total Environ.* **2021**, *774*, 144837. [\[CrossRef\]](#)
65. Liu, H.; Remme, R.P.; Hamel, P.; Nong, H.; Ren, H. Supply and demand assessment of urban recreation service and its implication for greenspace planning—A case study on Guangzhou. *Landsc. Urban Plan.* **2020**, *203*, 103898. [\[CrossRef\]](#)
66. Tianlin, Z.; Dou, Z.; Chenchen, Z. How to optimize ecological compensation to alleviate environmental injustice in different cities in the Yellow River Basin? A case of integrating ecosystem service supply, demand and flow. *Sustain. Cities Soc.* **2021**, *75*, 103341. [\[CrossRef\]](#)
67. Xin, G.; Juqin, S.; Weijun, H.; Xu, Z.; Zhichao, L.; Weifang, H.; Jingzhe, W.; Yingjie, R.; Xin, Z. Spatial-temporal analysis of ecosystem services value and research on ecological compensation in Taihu Lake Basin of Jiangsu Province in China from 2005 to 2018. *J. Clean. Prod.* **2021**, *317*, 128241. [\[CrossRef\]](#)
68. Su, D.; Cao, Y.; Dong, X.; Wu, Q.; Fang, X.; Cao, Y. Evaluation of ecosystem services budget based on ecosystem services flow: A case study of Hangzhou Bay area. *Appl. Geogr.* **2023**, *162*, 103150. [\[CrossRef\]](#)
69. Cortinovis, C.; Zulian, G.; Geneletti, D. Assessing Nature-Based Recreation to Support Urban Green Infrastructure Planning in Trento (Italy). *Land* **2018**, *7*, 112. [\[CrossRef\]](#)
70. Li, K.; Hou, Y.; Andersen, P.S.; Xin, R.; Rong, Y.; Skov-Petersen, H. An ecological perspective for understanding regional integration based on ecosystem service budgets, bundles, and flows: A case study of the Jinan metropolitan area in China. *J. Environ. Manag.* **2022**, *305*, 114371. [\[CrossRef\]](#)
71. Chen, J.; Xu, M.; Su, X.; Gao, J. Spatial transfer of regional ecosystem service in Nanjing City. *Acta Ecol. Sin.* **2014**, *34*, 5087–5095. [\[CrossRef\]](#)
72. Ma, Y.; Chen, H.; Yang, M.; Aihemaiti, G.; Lu, W.; Zhao, R. Ecological compensation based on multiscale ecosystem carbon sequestration service flow. *J. Environ. Manag.* **2024**, *372*, 123396. [\[CrossRef\]](#)
73. Chi, C.; Shen, J.; Gao, X.; Hu, P.; Yi, P. Horizontal ecological compensation standards based on ecosystem services flow. *Ecol. Indic.* **2024**, *163*, 112081. [\[CrossRef\]](#)
74. Yu, Y.; Li, J.; Han, L.; Zhang, S. Research on ecological compensation based on the supply and demand of ecosystem services in the Qinling-Daba Mountains. *Ecol. Indic.* **2023**, *154*, 110687. [\[CrossRef\]](#)
75. Wu, C.; Lu, R.; Zhang, P.; Dai, E. Multilevel ecological compensation policy design based on ecosystem service flow: A case study of carbon sequestration services in the Qinghai-Tibet Plateau. *Sci. Total Environ.* **2024**, *921*, 171093. [\[CrossRef\]](#)
76. Pinto, L.V.; Inacio, M.; Bogdzevic, K.; Kalinauskas, M.; Gomes, E.; Pereira, P. Factors affecting cultural ecosystem services use in Vilnius (Lithuania): A participatory mapping survey approach. *Heliyon* **2023**, *9*, e15384. [\[CrossRef\]](#)
77. Li, J.; Geneletti, D.; Wang, H. Understanding supply-demand mismatches in ecosystem services and interactive effects of drivers to support spatial planning in Tianjin metropolis, China. *Sci. Total Environ.* **2023**, *895*, 165067. [\[CrossRef\]](#)
78. Dang, H.; Li, J. Supply-demand relationship and spatial flow of urban cultural ecosystem services: The case of Shenzhen, China. *J. Clean. Prod.* **2023**, *423*, 138765. [\[CrossRef\]](#)
79. Zhao, Q.; Li, J.; Cuan, Y.; Zhou, Z. The Evolution Response of Ecosystem Cultural Services under Different Scenarios Based on System Dynamics. *Remote Sens.* **2020**, *12*, 418. [\[CrossRef\]](#)
80. Wu, J.; Huang, Y.; Jiang, W. Spatial matching and value transfer assessment of ecosystem services supply and demand in urban agglomerations: A case study of the Guangdong-Hong Kong-Macao Greater Bay area in China. *J. Clean. Prod.* **2022**, *375*, 134081. [\[CrossRef\]](#)
81. Clemente, P.; Calvache, M.; Antunes, P.; Santos, R.; Cerdeira, J.O.; Martins, M.J. Combining social media photographs and species distribution models to map cultural ecosystem services: The case of a Natural Park in Portugal. *Ecol. Indic.* **2019**, *96*, 59–68. [\[CrossRef\]](#)
82. Su, B.R.; Liu, M.C. An ecosystem service trade-off management framework based on key ecosystem services. *Ecol. Indic.* **2023**, *154*, 110894. [\[CrossRef\]](#)

83. Ament, J.M.; Moore, C.A.; Herbst, M.; Cumming, G.S. Cultural Ecosystem Services in Protected Areas: Understanding Bundles, Trade-Offs, and Synergies. *Conserv. Lett.* **2016**, *10*, 440–450. [\[CrossRef\]](#)
84. Cheng, X.; Van Damme, S.; Li, L.; Uyttenhove, P. Cultural ecosystem services in an urban park: Understanding bundles, trade-offs, and synergies. *Landsc. Ecol.* **2022**, *37*, 1693–1705. [\[CrossRef\]](#)
85. Yıldırım, G.; Akamca, G.Ö. The effect of outdoor learning activities on the development of preschool children. *S. Afr. J. Educ.* **2017**, *37*, 1–10. [\[CrossRef\]](#)
86. Dang, H.; Li, J.; Zhou, Z. Revealing the supply-demand relationship of urban cultural ecosystem services: The combination of open-source spatial model and topic model. *Appl. Geogr.* **2024**, *167*, 103288. [\[CrossRef\]](#)
87. Yin, D.; Huang, Q.; He, C.; Hua, X.; Liao, C.; Inostroza, L.; Zhang, L.; Bai, Y. The varying roles of ecosystem services in poverty alleviation among rural households in urbanizing watersheds. *Landsc. Ecol.* **2022**, *37*, 1673–1692. [\[CrossRef\]](#)
88. Chen, J.; Jiang, B.; Bai, Y.; Xu, X.; Alatalo, J.M. Quantifying ecosystem services supply and demand shortfalls and mismatches for management optimisation. *Sci. Total Environ.* **2019**, *650*, 1426–1439. [\[CrossRef\]](#) [\[PubMed\]](#)
89. Syrbe, R.-U.; Walz, U. Spatial indicators for the assessment of ecosystem services: Providing, benefiting and connecting areas and landscape metrics. *Ecol. Indic.* **2012**, *21*, 80–88. [\[CrossRef\]](#)
90. Huai, S.Y.; Chen, F.; Liu, S.; Canters, F.; van de Voorde, T. Using social media photos and computer vision to assess cultural ecosystem services and landscape features in urban parks. *Ecosyst. Serv.* **2022**, *57*, 101475. [\[CrossRef\]](#)
91. He, S.; Su, Y.; Shahtahmassebi, A.R.; Huang, L.; Zhou, M.; Gan, M.; Deng, J.; Zhao, G.; Wang, K. Assessing and mapping cultural ecosystem services supply, demand and flow of farmlands in the Hangzhou metropolitan area, China. *Sci. Total Environ.* **2019**, *692*, 756–768. [\[CrossRef\]](#)
92. Wang, Y.; Hayashi, K. Methodological development of cultural ecosystem services evaluation using location data. *J. Clean. Prod.* **2023**, *396*, 136523. [\[CrossRef\]](#)
93. Zhu, W.; Zhang, Z.; Zhang, H.; Tovvudorj, R. Integrating spatial patterns and driving factors of cultural ecosystem services into territorial spatial governance: A case study of the Horqin Sandy Land with multi-ethnic settlements. *Habitat Int.* **2024**, *148*, 103093. [\[CrossRef\]](#)
94. van Zanten, B.T.; Van Berkel, D.B.; Meentemeyer, R.K.; Smith, J.W.; Tieskens, K.F.; Verburg, P.H. Continental-scale quantification of landscape values using social media data. *Proc. Natl. Acad. Sci. USA* **2016**, *113*, 12974–12979. [\[CrossRef\]](#)

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